ASSIGNMENT 02
Total Marks: 100
Unique no.:202585

ONLY FOR YEAR MODULE

All questions will be marked.

DO NOT USE A CALCULATOR TO OBTAIN YOUR ANSWERS-WHERE APPLICABLE, LEAVE YOUR ANSWERS IN TERMS OF FACTORIALS, $_nC_r$ AND $_nP_r$.

Question 1: **8 Marks**

Suppose $A = \{a, b, c, d, e, f\}$ and R is the relation on A defined by

$$R = \{(a, f), (b, a), (c, d), (d, d), (e, b), (f, e)\}.$$

State, with reason, whether the relation is:

(1.1) A function from A to A. (2)

(1.2) An everywhere defined function. (2)

(1.3) An onto function. (2)

(1.4) A one-to-one function. (2)

Question 2: **10 Marks**

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the function defined by $f(x, y) = (x + y, x - y)$. Show that

(2.1) f is one-to-one. (4)

(2.2) Find f^{-1}. Show that $\left(f \circ f^{-1}\right)(x, y) = (x, y)$. (6)

Question 3: 16 Marks

(3.1) Use the definition of $O(f)$ to show that n^2 is $O(n^2 \log n)$. (10)

(3.2) Is $n^2 \log n$ also $O(n^2)$? Prove your answer. (6)

Question 4: 16 Marks

Use the rules (and if necessary, the definition) for ordering Θ-classes to arrange the following in order from lowest to highest:

$$\lg n; \quad (1,0001)^n; \quad \lg n^2; \quad (\lg n)^2; \quad n^3 + 4; \quad 24n^3 + 16n; \quad n^2 + 5n.$$

Question 5: 6 Marks

Let A have 5 elements and B have 4 elements.

(5.1) How many everywhere defined functions are there from A to B? (4)

(5.2) How many everywhere defined one-to-one functions are there from A to B? Explain your answer. (2)

Question 6: 34 Marks

Let

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 8 & 3 & 5 & 1 & 2 & 7 & 6 \end{pmatrix}.$$

(6.1) Write p as a function, e.g. $p(1) = 3, p(2) = \ldots$. (2)

(6.2) Write p as a product of disjoint cycles. (3)

(6.3) Write p as a product of transpositions. (3)

(6.4) Is p an even or odd permutation? (3)

(6.5) Compute $p \circ p$. (4)

(6.6) Compute p^{-1}. (4)

(6.7) Determine the period of p, that is, the smallest positive integer k such that (6)

$$p^k = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{pmatrix}.$$

(6.8) Find a permutation q such that (9)

$$q \circ p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 4 & 3 & 1 & 8 & 6 & 5 & 7 \end{pmatrix}.$$

Question 7: **4 Marks**

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Compute the product (composition)

$$(1, 3, 8) \circ (3, 5, 7, 8) \circ (2, 4)$$

and write it in the form

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ & & & & & & & \end{pmatrix}.$$

Question 8: **6 Marks**

(8.1) Let $A = \{a, b, c, d, e, f, g\}$. Write the following permutation as a product of disjoint cycles. (2)

$$\begin{pmatrix} a & b & c & d & e & f & g \\ d & c & f & a & g & b & e \end{pmatrix}.$$

(8.2) Determine whether the following permutation is even or odd. (4)

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 6 & 1 & 5 & 7 & 2 & 4 & 3 \end{pmatrix}.$$